

Homework 6

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*All of my homework is available at <https://github.com/lacampanella233/Homework.git>.

Problem 1 (3-8). If $\sum a_n$ converges and $\{b_n\}$ is monotone and bounded, prove that

$$\sum a_n b_n$$

also converges.

Proof. Denote $s_n = a_0 + a_1 + \cdots + a_n$. By the Abel summation formula, we have

$$S_n := \sum_{k=1}^n a_k b_k = s_n b_n + \sum_{k=1}^{n-1} s_k (b_k - b_{k+1}).$$

We will prove that $\sum a_n b_n$ is a Cauchy series, hence $\sum a_n b_n$ converges. Note that $\{s_n\}$ and $\{b_n\}$ all converge, thus $\{s_n b_n\}$ also converges, and there exist $M > 0$ such that $|s_n| < M$ for all $n \in \mathbb{N}_+$.

For any positive number ε , there exist $N_1, N_2 \in \mathbb{N}_+$ such that

$$|s_m b_m - s_n b_n| < \frac{\varepsilon}{2}, \forall m > n > N_1; \quad |b_m - b_n| < \frac{\varepsilon}{2M}, \forall m > n > N_2.$$

Let $N = \max\{N_1, N_2\}$. For any $m > n > N$, then for all $m > n > N$,

$$\begin{aligned} |S_m - S_n| &= \left| s_m b_m - s_n b_n + \sum_{k=n}^{m-1} s_k (b_k - b_{k+1}) \right| \\ &\leq |s_m b_m - s_n b_n| + \sum_{k=n}^{m-1} |s_k| \cdot |b_k - b_{k+1}| < \frac{\varepsilon}{2} + M \sum_{k=n}^{m-1} |b_k - b_{k+1}| \\ &= \frac{\varepsilon}{2} + M |b_{n+1} - b_m| \quad (\text{This is because } \{b_n\} \text{ is monotone}) \\ &< \varepsilon. \end{aligned}$$

Therefore, $\sum a_n b_n$ is a Cauchy series, and $\sum a_n b_n$ converges. □

Problem 2 (3-9). Find the radius of convergence of each of the following power series: (a) $\sum n^3 z^n$, (b) $\sum \frac{2^n}{n!} z^n$, (c) $\sum \frac{2^n}{n^2} z^n$, (d) $\sum \frac{n^3}{3^n} z^n$.

Solution. (a)

$$R = \frac{1}{\limsup \sqrt[n]{n^3}} = \frac{1}{1} = 1.$$

(b)

$$R = \frac{1}{\limsup \sqrt[n]{2^n/n!}} = \frac{1}{0} = \infty.$$

(c)

$$R = \frac{1}{\limsup \sqrt[n]{2^n/n^2}} = \frac{1}{2 \limsup n^{-2/n}} = \frac{1}{2}.$$

(d)

$$R = \frac{1}{\limsup \sqrt[n]{n^3/3^n}} = \frac{1}{3 \limsup n^{-3/n}} = \frac{1}{3}.$$

Problem 3 (3-10). Assume that all coefficients of the power series

$$\sum a_n z^n$$

are integers, and infinitely many of them are nonzero. Prove that its radius of convergence is at most 1.

Proof. It suffices to show that $\limsup_{n \rightarrow \infty} \sqrt[n]{a_n} \geq 1$. For any positive integer N , there exists $n > N$ such that $a_n \geq 1$ (since infinitely many of the coefficients are nonzero), then the conclusion holds. $\square$ **Problem 4 (3-14).** Let $\{s_n\}$ be a sequence of complex numbers, and define its arithmetic means by

$$\sigma_n = \frac{s_0 + s_1 + \cdots + s_n}{n+1}, \quad (n = 0, 1, 2, \dots).$$

- (a) If $\lim_{n \rightarrow \infty} s_n = s$, prove that $\lim_{n \rightarrow \infty} \sigma_n = s$.
- (b) Construct a sequence $\{s_n\}$ such that $\lim_{n \rightarrow \infty} \sigma_n = 0$ but $\{s_n\}$ does not converge.
- (c) Can the following happen: $s_n > 0$ for all n , $\lim_{n \rightarrow \infty} \sigma_n = 0$, but $\limsup_{n \rightarrow \infty} s_n = \infty$?
- (d) For $n \geq 1$, let $a_n = s_n - s_{n-1}$. Prove that

$$s_n - \sigma_n = \frac{1}{n+1} \sum_{k=1}^n k a_k.$$

Assume further that $\lim_{n \rightarrow \infty} (n a_n) = 0$ and that $\{\sigma_n\}$ converges. Prove that $\{s_n\}$ converges. (This is a partial converse of (a), with the additional assumption $n a_n \rightarrow 0$.)

- (e) Derive the conclusion in (d) under weaker hypotheses: assume $M < \infty$, $|n a_n| \leq M$ for all n , and $\lim_{n \rightarrow \infty} \sigma_n = \sigma$. Prove $\lim_{n \rightarrow \infty} s_n = \sigma$ by following these steps:

If $m < n$, then

$$s_n - \sigma_n = \frac{m+1}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{i=m+1}^n (s_n - s_i).$$

For such i ,

$$|s_n - s_i| \leq \frac{(n-i)M}{i+1} \leq \frac{(n-m-1)M}{m+2}.$$

Fix $\varepsilon > 0$. For each n , choose a positive integer m such that

$$m \leq \frac{n - \varepsilon}{1 + \varepsilon} < m + 1.$$

Then

$$\frac{m + 1}{n - m} \leq \frac{1}{\varepsilon},$$

and $|s_n - s_i| < M\varepsilon$. Hence

$$\limsup_{n \rightarrow \infty} |s_n - \sigma| \leq M\varepsilon.$$

Since ε is arbitrary, conclude that $\lim_{n \rightarrow \infty} s_n = \sigma$.

Proof. (a) For any positive number ε , there exists $N_1 \in \mathbb{N}_+$ such that $|s_n - s| < \varepsilon/3$ for all $n > N_1$. Take $N_2 \in \mathbb{N}_+$ such that $s_0 + s_1 + \cdots + s_{N_1} < (\varepsilon/3)N_2$. It's clear that there exists $N_3 \in \mathbb{N}_+$ such that $(N_1 + 1)s < N_3\varepsilon/3$.

Set $N = \max\{N_1, N_2, N_3\}$. For any $n > N$,

$$\begin{aligned} |\sigma_n - s| &\leq \left| \frac{s_0 + s_1 + \cdots + s_{N_1}}{n} \right| + \left| \frac{s_{N_1+1} + s_{N_1+2} + \cdots + s_n}{n} - s \right| \\ &\leq \frac{\varepsilon}{3} + \left| \frac{(n - N_1)s}{n} - s \right| + \frac{|s_{N_1+1} - s| + |s_{N_1+2} - s| + \cdots + |s_n - s|}{n} \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence $\lim_{n \rightarrow \infty} \sigma_n = s$.

(b) Let $s_n = (-1)^n$. Then $\sigma_n = \frac{1}{n+1} \sum_{k=0}^n (-1)^k$. It's easy to verify that $\lim_{n \rightarrow \infty} \sigma_n = 0$, but $\{s_n\}$ does not converge.

(c) Yes. Let

$$s_n = \begin{cases} k, & \text{if } n = 2^k; \\ 2^{-n}, & \text{otherwise.} \end{cases}$$

It's not hard to verify that $s_n = \mathcal{O}(\log n)$, then $\lim_{n \rightarrow \infty} \sigma_n = 0$. But by the definition of s_n , we have $\limsup_{n \rightarrow \infty} s_n = +\infty$.

(d) Note that

$$\begin{aligned} \sum_{k=1}^n k a_k &= \sum_{k=1}^n k(s_k - s_{k-1}) = \sum_{k=1}^n k s_k - \sum_{k=1}^n k s_{k-1} = n s_n + \sum_{k=1}^{n-1} k s_k - \sum_{k=0}^{n-1} (k+1) s_k \\ &= n s_n - \sum_{k=0}^{n-1} s_k = (n+1)(s_n - \sigma_n). \end{aligned}$$

Suppose $\lim_{n \rightarrow \infty} n a_n = 0$ and $\{\sigma_n\}$ converges. We only need to prove that $\lim_{n \rightarrow \infty} (s_n - \sigma_n) = 0$. This is an immediate consequence of (a), with substituting s_n to $n a_n$.

(e) Assume $M < \infty$, $|n a_n| \leq M$ for all n , and $\sigma_n \rightarrow \sigma$. For $m < n$, the identity in the statement gives

$$s_n - \sigma_n = \frac{m+1}{n-m}(\sigma_n - \sigma_m) + \frac{1}{n-m} \sum_{i=m+1}^n (s_n - s_i).$$

For each $i \in \{m+1, \dots, n\}$, $s_n - s_i = \sum_{k=i+1}^n a_k$, hence

$$|s_n - s_i| \leq \sum_{k=i+1}^n |a_k| \leq \sum_{k=i+1}^n \frac{M}{k} \leq \frac{(n-i)M}{i+1} \leq \frac{(n-m-1)M}{m+2}.$$

Fix $\varepsilon > 0$. For each n , choose $m \in \mathbb{N}_+$ such that

$$m \leq \frac{n-\varepsilon}{1+\varepsilon} < m+1.$$

Then $n-m \geq \varepsilon(m+1) + \varepsilon > \varepsilon(m+1)$, so

$$\frac{m+1}{n-m} \leq \frac{1}{\varepsilon}.$$

Also, $n-m-1 < \varepsilon(m+2)$, therefore for $i = m+1, \dots, n$,

$$|s_n - s_i| \leq \frac{(n-m-1)M}{m+2} < M\varepsilon.$$

Taking absolute values in the decomposition of $s_n - \sigma_n$, we get

$$|s_n - \sigma_n| \leq \frac{m+1}{n-m} |\sigma_n - \sigma_m| + \frac{1}{n-m} \sum_{i=m+1}^n |s_n - s_i| \leq \frac{1}{\varepsilon} |\sigma_n - \sigma_m| + M\varepsilon.$$

Since $m \rightarrow \infty$ as $n \rightarrow \infty$ and $\sigma_n \rightarrow \sigma$, we have $|\sigma_n - \sigma_m| \rightarrow 0$, hence $\limsup_{n \rightarrow \infty} |s_n - \sigma_n| \leq M\varepsilon$.

Finally, $|s_n - \sigma| \leq |s_n - \sigma_n| + |\sigma_n - \sigma|$, and $\sigma_n \rightarrow \sigma$ implies

$$\limsup_{n \rightarrow \infty} |s_n - \sigma| \leq M\varepsilon.$$

Because $\varepsilon > 0$ is arbitrary, $\lim_{n \rightarrow \infty} s_n = \sigma$. □